



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

BDCAC401

BASICS OF AUTOCAD

Apply Basics of AutoCAD

Competence

RQF Level: 4

Learning Hours



Credits: 6

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: BDCAC401- TVET Certificate 4 In Building Construction

Copyright: © Rwanda TVET Board, 2023

Issue Date: April, 2023

Purpose statement	This module describes the knowledge, skills and attitude required to create highly precise 2D designs that visualize construction drawings. It is intended to be delivered to learners pursuing TVET level IV in building construction. At the end of this module, the learner will be able to create, modify and plot construction drawings.					
Leaning assumed to be in place	Advanced construction drawing					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	Group project and presentation	30%				
	Individual project /Work	70%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Describe AutoCAD software	1.1. AutoCAD software is adequately described as per developer
	1.2. AutoCAD software is properly differentiated from other CAD Software
	1.3. AutoCAD software is properly installed as per installation user guide
	1.4. AutoCAD user interface is appropriately Explored according to chosen workspace
	1.5. Basic operations are perfectly applied in AutoCAD
2. Create Drawings	2.1. AutoCAD workspace is correctly customized / Created according to the given work
	2.2. Layers are properly created and named as per types of drawings
	2.3. Construction drawings are perfectly created using coordinates
	2.4. Construction drawings are perfectly created using AutoCAD Drawing tools/Commands
	2.5. Construction drawings are correctly modified using AutoCAD modifying commands
	2.6. Hatches and fills are correctly applied as per drawings requirements
	2.7. Blocks are correctly created and used/inserted as per drawings or object types
	2.8. Annotation tools are correctly used according to the types of notes and dimensions to be added
	2.9. Title block is appropriately created according to the use of the drawings
3. Plot drawings	3.1. Working with paper space is correctly done as per types and requirements of the drawings
	3.2. Advanced plotting options are correctly explored according to drawings types and

	sizes
	3.3. Scales are correctly set/chosen according to drawings types and sizes
	3.4. Construction drawings are correctly printed/Plotted as per their sizes

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Identify Units of measurement ✓ Identify drawing tools and equipment ✓ Interpret drawing terminologies ✓ Interpret drawing Symbols ✓ Identify drawing paper format ✓ Identify annotations and labelling ✓ Describe the geometrical shapes ✓ Identify AutoCAD installation requirements ✓ Identify AutoCAD tools ✓ Identify layout	✓ Use drawing tools and Equipment ✓ Apply basic ICT skills ✓ Take measurement of objects ✓ Install AutoCAD software ✓ Perform AutoCAD basic settings ✓ Create geometrical shapes/Drawings ✓ Apply dimensions and text ✓ Create layout ✓ Perform Printing	✓ Use creativity and innovation throughout the design works ✓ Pay attention to design details ✓ Demonstrate resourcefulness in the new design trends ✓ Demonstrate persistent, team working, flexibility, punctuality during project design

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Describe AutoCAD software
-------------------	---

		2. Create drawings 3. Plot drawings
Learning outcome 1: Describe AutoCAD software		Learning hours: 5
Indicative content		
• Description of AutoCAD software ➤ CAD and history of CAD ➤ AutoCAD and history AutoCAD ➤ AutoCAD uses and users ➤ Benefits of using AutoCAD software • AutoCAD software Installation ➤ AutoCAD installation requirements ➤ AutoCAD licensing ➤ AutoCAD installation process • Exploring AutoCAD user interface ➤ Starting AutoCAD ➤ AutoCAD default workspaces ➤ Keyboard and mouse functions ➤ Opening or accessing an existing drawing ➤ Application menu ➤ Elements of the menu bar ➤ Elements of quick access Toolbar ➤ Ribbon ➤ Elements of the status bar ➤ File tabs ➤ Layout tabs ➤ Toolbars ➤ Command line • Apply Basic operations in AutoCAD ➤ Create new drawing ➤ Open an existing drawing ➤ Save/Save as ➤ Undo/Redo ➤ Pan ➤ Zoom ➤ AutoCAD file formats		
Resources required for the learning outcome		
Equipment	- Computer laboratory, projector, flip chart, mark pens, Board, Electricity, Electrical cables	

Materials	- Eraser, drawing papers
Instrument	- Ruler, ink-pen, pencils, pencil sharpener
Facilitation techniques	• Group discussion • Trainer guided • Practical exercises
Formative assessment methods	• Written assessment • Oral assessment • Practical assessment • Matching questions • Multiple choice • Short questions

Learning outcome 2: Create Construction Drawings	Learning hours: 45
Indicative content	
• Workspace Customization / creation ➤ 2D Drafting and annotation workspace customization ➤ Display specific menus; ➤ Display specific toolbars; ➤ Display specific palettes; ➤ Display specific ribbon panels. ➤ New workspace creation ➤ Display specific menus, toolbars, palettes, and panels • Working with Layers ➤ Creation of layers ➤ Naming/renaming layers ➤ Modification of layer properties ➤ Freezing/Thawing layers ➤ Changing layer colours ➤ Working with a specific layer • Create drawings with coordinates ➤ Drawing with absolute Coordinates ➤ Drawing with relative coordinates • Create Construction Drawings are perfectly created using AutoCAD Drawing tools/Commands ➤ Line, Polyline, Erase, Rectangle, Circle, Arc, Polygon, Spline, Table, etc... ➤ Drawing limits ➤ Apply AutoCAD status bar toggles (Dynamic Input, Polar Tracking, etc...) • Apply Editing/modifying commands ➤ Offset, Trim, Extend, Stretch, Fillet, Rotate, Mirror, Move, Copy, Array, Scale, Explode, Break • Apply Hatches and Fills	

- The BHATCH Command
- Solid Hatches
- Editing properties of hatches
- The SKETCH Command
- Gradients
- **Work with Blocks**
 - Insert a Bloc
 - Create blocks within a Drawing
 - Modify a block
 - Group objects
 - Create Block Libraries
- **Annotation**
 - Changing annotative settings
 - Work with annotative Object Scale dialog
 - Set annotation scale
 - Add dimensions with dimensions pallet tools
 - Add dimensions with TEXT pallet tools
 - Add dimensions and text using command/Command line
- **Create Title block**
 - Identification of Elements of the title bloc
 - Title block templates
 - Title block creation

Resources required for the indicative content

Equipment	- Computer laboratory, projector, flip chart, mark pens, Board, Electricity, Electrical cables
Materials	- Eraser, drawing papers
Instrument	- Ruler, ink-pen, pencils, pencil sharpener
Facilitation techniques	• Group discussion • Trainer guided • Practical exercises
Formative assessment methods	• Written assessment • Oral assessment • Practical assessment

Learning outcome 3: Plot drawings

Learning hours:10

Indicative content

- **Working with paper space**
 - Page setup for layout
 - Making and clipping viewports

- Setting scale of objects in viewports
- Managing layer visibility in viewports
- Inserting a title block and ruler in layout
- Plotting from model space with proper scale
- Plotting a drawing from layout
- **Advanced plotting options**
 - Making and using color in plot style table
 - Making and using named plot style table
 - Understanding Annotative property
 - Making annotative objects
 - Markup set manager
 - Detail look at plot stamp
- **Other plotting options**
 - Publishing sheet set
 - Importing page setup
 - Final Notes

Resources required for the indicative content

Equipment	- computer laboratory, projector, flip chart, mark pens, board, electricity, electrical cables
Materials	- eraser, drawing papers
Instrument	- ruler, ink-pen, pencils, pencil sharpener
Facilitation techniques	• Group discussion • Trainer guided • Practical exercises
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment • Project based assessment

Integrated/Summative assessment

Integrated situation

Mr. KABERA, a resident of Gicumbi District, Rukomo center planned to build a residential house for his family 2 months ago. For this purpose, his son who is a final year student at IPRC Kigali provided him with both soft (pdf) and hard documents for that project. Unfortunately, soft documents get lost, and Mr. KABERA left only with hard documents that Include site plan, floor plan, foundation plan, elevation views, roof plan and both longitudinal sections and cross-sections.

Using AutoCAD software, you are required to propose the title block with all required information for better project understanding and using a scale of your choice, reproduce: The site plan, Floor plan, elevation views, and roof plan, and both longitudinal sections and cross-sections on an A0 paper in pdf format in 3 days so that Mr. KABERA could apply for construction permit.

Resources

Equipment	- Computer, electricity
Materials	- Hard documents (drawings)
Instrument	NA

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 2: Create construction Drawings (90%)	2.1. Workspace is well customized / created	Workspace chosen			2
		Specific menus displayed			2
		Specific toolbars displayed			2
		Specific palettes displayed			2
		Specific ribbon panels displayed			2
	2.2. Layers are Properly created	Layers created			2
		Layers named			2
		Layer properties modified			2
		Layers Frozen/ thawed			2
		Layer colors customized			2
		Layer set current			2
	2.3. Construction Drawings are perfectly	Absolute Coordinates			2

	created with coordinates	Relative Coordinates			2
	2.4. Construction Drawings are perfectly created with drawing tools/commands	Lines used			2
		Polylines used			2
		Rectangles used			2
		Circles used			2
		Arcs used			2
		Polygons used			2
		Splines used			2
		Tables used			2
	2.5. Construction Drawings are correctly modified	Offset tool used			1
		Trim tool used			1
		Extend tool used			1
		Stretch tool used			1
		Fillet tool used			1
		Rotate tool used			1
		Mirror tool used			1
		Move tool used			1
		Copy tool used			1
		Array tool used			1
		Scale tool used			1
		Explode tool used			1
		Break tool used			1
	2.6. Hatch and gradient patterns are correctly used	Hatch/gradient type applied			2
		Hatch/gradient pattern applied			2
		Hatch/gradient angle twisted			2
		Hatch/gradient color customized			2
		Hatch/gradient scale adjusted			2
	2.7. Blocks are correctly created	Blocks created and saved			5
	2.8. Annotation tools are correctly used	Texts applied			2
		Dimensions applied			2
		Leaders used			2
		Label used			2
		Center line applied			2
	2.9. Title block is appropriately created	Drawing number indicated			1

		Part names indicated			1
		Name of the firm indicated			1
		Scale indicated			1
		Date indicated			1
		Name of the draftsmen indicated			1
		Name of the checker indicated			1
		Name of the Person approving the drawing indicated			1
		The firm's logo applied			1
		Drawing sheet size indicated			1
Learning outcome 3: Plot drawings (10%)	3.1. Drawings are correctly plotted	Drawing Web Format (DWF) file created			2
		Portable Document Format (PDF) created			2
		JPG/PNG image created			2
		Paper size (A0) respected			2
		Margins drawn			2
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References:

1. AutoCAD basics. (2020). In *Construction Detailing for Interior Design*.
<https://doi.org/10.5040/9781501326431.ch-003>
2. Butorina, I. V., & Vasilieva, N. (2018). Parametric modeling in architectural and construction design in AutoCAD. *IOP Conference Series: Materials Science and Engineering*.
<https://doi.org/10.1088/1757-899X/451/1/012121>
3. Gindis, E. J., & Kaebisch, R. C. (2022). AutoCAD fundamentals—Part II. In *Up and Running with AutoCAD® 2022*. <https://doi.org/10.1016/b978-0-323-89923-9.00002-0>
4. Irawan M. R, I., & H.M, A. (2017). A CASE STUDY OF AUTOCAD 2D ENGINEERING DRAWING PERFORMANCE AMONG FURNITURE AND PRODUCT DESIGN (BFPD) STUDENTS.
<https://doi.org/10.17501/icedu.2017.3106>
5. Onstott, S. (2017). AutoCAD® and AutoCAD LT® Essentials. In *AutoCAD® and AutoCAD LT® Essentials*. <https://doi.org/10.1002/9781119414124>
6. Sadeghpour, F., Moselhi, O., & Alkass, S. T. (2006). Computer-Aided Site Layout Planning. *Journal of Construction Engineering and Management*. [https://doi.org/10.1061/\(asce\)0733-9364\(2006\)132:2\(143\)](https://doi.org/10.1061/(asce)0733-9364(2006)132:2(143))
7. Saputro, B. E., & Supratno, S. (2020). Implementation of the use of AutoCAD applications in increasing basic competence in describing techniques in vocational education. *IOP Conference Series: Materials Science and Engineering*. <https://doi.org/10.1088/1757-899X/830/4/042022>
8. Up and Running with AutoCAD® 2022. (2022). In *Up and Running with AutoCAD® 2022*.
<https://doi.org/10.1016/c2020-0-03148-8>
9. Zhao, Z., & Zhu, X. (2019). Automatic drawing technique for horizontal projection diagrams of exploration borehole deviations. *Multimedia Tools and Applications*.
<https://doi.org/10.1007/s11042-018-6659-0>
10. Ziden, A. A., Zakaria, F., & Othman, A. N. (2012). Effectiveness of AutoCAD 3D Software as a learning support tool. *International Journal of Emerging Technologies in Learning*.
<https://doi.org/10.3991/ijet.v7i2.1906>